AURA Twin

History + wearable state.
Fresh-patient evidence.
A challengeable prediction.

AURA-SYNTHETIC-2.0.0

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Generated data only · No clinical validation or treatment recommendation

Freeze the prediction. Then reveal the outcome.

Current model aura-synthetic-2.0.0 · synthetic benchmark evidence

01

Choose

Patient, cutoff, sensor case.
Copy and truncate history.

02

Freeze

Predict before seeing outcomes.
Preserve output + hashes.

03

Reveal

Original recorded future.
No recomputation.

04

Compare

Model vs persistence.
Interval hits + receipt.

Public data: this enforces a computation boundary, not secure study blinding.

Four sensor cases expose different failure modes.

Current model aura-synthetic-2.0.0 · synthetic benchmark evidence

Clean replay

An honest baseline; a local window may lose to persistence.

10-minute CGM gap

Forward fill permitted; show degraded quality.

30-minute dropout

Stale signal; withhold forecast and probability.

Undetected sensor bias

Plausible +30 mg/dL input shift may pass freshness checks.

The reference future stays unchanged. Persistence sees the same perturbed current value.

Compare models on the same fresh routine cases.

Current model aura-synthetic-2.0.0 · synthetic benchmark evidence

Routine	Current MAE	Legacy MAE	Persistence	Band cover
Regular	10.06	10.88	21.68	92.28%
Meals +2h	9.72	21.92	21.47	92.84%
Meals +4h	9.29	22.92	19.57	93.34%
Jittered meals	13.97	17.05	21.11	85.31%
Daily changes + skips	13.12	17.64	19.45	86.75%
Variable sleep/activity	13.57	19.39	18.65	86.01%

All rows use the same fresh patient identities. Routine families were declared during development.

Make the experiment reproducible offline.

Current model aura-synthetic-2.0.0 · synthetic benchmark evidence

01

What can be checked

Raw model bytes → SHA-256
Canonical input prefix → SHA-256
Recomputed forecast and revealed score
Mismatch detection for altered artifacts

02

What is not established

No digital signature
No independent timestamp
No trusted authorship guarantee
No clinical audit certification

```
node scripts/verify_receipt.mjs path/to/receipt.json
```

A forecast you can inspect.

A working fusion pipeline.
A measured synthetic improvement.
A reproducible challenge.

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